INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR
End Semester Examination 2024-25

Date of Examination: /11/24 Session(FN/AN) _ Duration 3 hrs, Marks = 80
Sub No: CS60075 Sub Name: Natural Language Processing
Department/Centre/School: Computer Science and Engineering
Specific charts, graph paper, log book etc. required NO
Special Instructions (if any) ANSWER ALL questions. **No clarifications!** In case of reasonable doubt, make assumptions and state them upfront. Marks will be deducted for sketchy answers and claims without proper reasoning. **All parts of a single question should be done at the same place.**

1. RNNs: Consider a simple RNN cell with parameters:

- Input-to-hidden weight $W_x = 0.2$
- Hidden-to-hidden weight $W_h = 0.5$
- Output weight $W_y = 1.0$
- Bias $b = 0.0$

The hidden state is computed as: $h_t = \tanh(W_x \cdot x_t + W_h \cdot h_{t-1} + b)$

The output is computed as: $y_t = W_y \cdot h_t$

Given:

- Inputs $x_1 = 0.4$ and $x_2 = 0.6$
- Target outputs $y_{\text{target},1} = 0.3$ and $y_{\text{target},2} = 0.5$
- Initial hidden state $h_0 = 0$
- Mean Squared Error (MSE) loss: Loss $= \frac{1}{2}\sum_t (y_t - y_{\text{target},t})^2$

(a) Perform the forward pass and calculate h_1, h_2, y_1, and y_2.

(b) Calculate the loss.

(c) Using BPTT, compute the gradients of the loss with respect to y_1, y_2, h_1 and h_2.

[6 + 2 + (2 × 3 + 3) = 17]

2. **Scaled dot-product attention**: Given the following matrices for query Q, key K, and value V:

$$Q = \begin{bmatrix} 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}, \quad K = \begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 1 \end{bmatrix}, \quad V = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$$

with $d_k = 3$, compute the output of the scaled dot-product attention mechanism as follows:

(a) Compute the attention scores by taking the dot product between Q and K^T, then divide by $\sqrt{d_k}$.

(b) Apply the softmax function to the attention scores to normalize them.

(c) Compute the weighted sum of values V using the attention scores obtained from (b).

[3 + 3 + 3 = 9]

3. Masked attention in sequence generation: In a sequence generation task, masked attention is applied to ensure that each position in the sequence only attends to itself and previous positions. Given a scenario where a sequence length is $n = 5$ and each token embedding has a dimensionality of $d = 4$, answer the following:

   (a) Describe why masked attention is critical for generating sequences in the decoder self-attention block.

   (b) For a sequence with tokens A, B, C, D, E, construct the mask matrix M for the decoder self-attention block.

   (c) Given a simplified attention score matrix (before softmax) as:

$$\text{Scores} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 1 & 1 & 1 & 0 & 0 \\ 1 & 1 & 1 & 1 & 0 \\ 1 & 1 & 1 & 1 & 1 \end{bmatrix}$$

   Apply the mask and normalize each row with the softmax function.

[2 + 2 + 6 = 10]

4. **LSTM-based prediction**: You are working with a dataset that contains daily temperatures (in degrees Celsius) recorded over 10 days as in the table below:

Day	Temperature
1	20
2	22
3	21
4	23
5	24
6	26
7	25
8	27
9	29
10	30

   You want to predict the temperature for Day 11 using an LSTM network. To do this, you will create sequences of length 3 (i.e., you will use the temperatures of the last 3 days to predict the next day).

   (a) Create the input-output pairs for the LSTM model based on the given data.

   (b) If the LSTM model predicts a temperature of 28 degrees Celsius for Day 11, what is the Mean Squared Error (MSE) when compared to the actual temperature of 29 degrees Celsius?

[6]

5. **Language modeling and smoothing**: You are developing a bigram language model based on the following small corpus of sentences:

   - `The cat sat on the mat.`
   - `The dog sat on the log.`
   - `Cats and dogs are friends.`
   - `The cat chased the dog.`

   Vocabulary size ($|V|$): 8 (`The, cat, sat, on, the, mat, dog, chased, and, are, friends`)

(a) Compute bigram probabilities (a) (`the, dog`) and (b) (`cat, chased`) using Laplace smoothing.

(b) If you encounter the bigram (`dogs, are`), what will its probability be using Laplace smoothing?

[9]

6. **Byte Pair Encoding (BPE)**: Given the following frequency table of English words:

Word	Frequency
low	5
lower	2
lowest	6
new	3
newer	2
newest	4

(a) In the above table add a new column: word sequence. In this coulumn write down each word as a sequence of characters separated by spaces. For example, the word `'low'` would be written as `'l o w'`.

(b) Using this sequence representation, apply BPE for **three iterations.** In each iteration **(a)** compute the frequencies of the byte pairs and find the pair to be merged, **(b)** update the sequences in the table accordingly (show the table with the updated sequences after each iteration). **Note:** For instance after the first iteration if `'o w'` is merged to form `'ow'` then the word sequence for the word `'low'` will change to `'l ow'`.

[1 + 9 = 10]

7. **Skip-grams**: Given the word vectors

$$V(w) = [0.5, 0.6, 0.2]$$
$$V(c_1) = [0.3, 0.8, 0.5]$$
$$V(c_2) = [0.9, 0.1, 0.4]$$

Compute $P(c|w)$ using the skip-gram model where w is the center word and c_1 and c_2 are the context words. Assume that cosine similarity expresses the similarity scores between a pair of word vectors and the standard way to convert similarity scores to probabilities is using softmax. [8]

8. **Beam search**: Consider a RNN decoder that generates sequences of tokens (e.g., words) from a given input. The model uses a vocabulary of size $V = 5$ with the following tokens:

- 0: "START"
- 1: "A"
- 2: "B"
- 3: "C"
- 4: "END"

When decoding a particular input, the RNN produces the following probabilities for the next tokens at each time step:

- **Time step 1**

$$P(0) = 0.0 \quad \text{(START)}$$
$$P(1) = 0.2 \quad \text{(A)}$$
$$P(2) = 0.5 \quad \text{(B)}$$
$$P(3) = 0.3 \quad \text{(C)}$$

- **Time step 2:**
  From "A":

$$P(0) = 0.0$$
$$P(1) = 0.1 \quad (A)$$
$$P(2) = 0.6 \quad (B)$$
$$P(3) = 0.3 \quad (C)$$

  From "B":

$$P(0) = 0.0$$
$$P(1) = 0.4 \quad (A)$$
$$P(2) = 0.4 \quad (B)$$
$$P(3) = 0.2 \quad (C)$$

  From "C":

$$P(0) = 0.0$$
$$P(1) = 0.3 \quad (A)$$
$$P(2) = 0.3 \quad (B)$$
$$P(3) = 0.4 \quad (C)$$

- **Time step 3**
  From "A":

$$P(0) = 0.5 \quad (\text{END})$$
$$P(1) = 0.0$$
$$P(2) = 0.3 \quad (B)$$
$$P(3) = 0.2 \quad (C)$$

  From "B":

$$P(0) = 0.0$$
$$P(1) = 0.5 \quad (A)$$
$$P(2) = 0.2 \quad (B)$$
$$P(3) = 0.3 \quad (C)$$

  From "C":

$$P(0) = 0.1 \quad (\text{END})$$
$$P(1) = 0.2 \quad (A)$$
$$P(2) = 0.5 \quad (B)$$
$$P(3) = 0.2 \quad (C)$$

(a) Using a beam width of $k = 2$, perform beam search decoding for the first **three time steps.** Show the beam at each step and the cumulative probabilities for each sequence. Note: For calculations you can just use product of probabilities directly rather than the sum of log probabilities as these products are not very small.

(b) Identify the two highest-probability sequences generated by the beam search at the end of **three time steps.**

[9 + 2 = 11]